

```
import networkx as nx
from scipy.stats import entropy
import itertools
```

```
def simple_phi(network, state):
    nodes = list(network.nodes)
    phi_candidates = []
```

```
    # Перебираем все partitions (только для
    # малых сетей!)
```

```
    for partition_size in range(1, len(nodes)//2
    + 1):
```

```
        for part1 in
        itertools.combinations(nodes, partition_size):
            part2 = [n for n in nodes if n not in
            part1]
```

```
            I_whole = mutual_info(network, state)
            I_parts =
            mutual_info_partition(network, part1, state) +
            \
            mutual_info_partition(network,
```

```
part2, state)
```

```
    phi_candidates.append(l_whole -  
l_parts)
```

```
    return min(phi_candidates, default=0)
```